CSA1708-ARTIFICIAL INTELLIGENCE

LAB EXPERIMENTS

Program 4: write the python program for cript arithmetric problem

Aim:

To write a Python program to solve the cryptarithmetic problem **SEND + MORE = MONEY** by assigning unique digits to each letter.

Algorithm:

1. Start the program.
2. Import the permutations function from the itertools module.
3. Generate all possible assignments of eight distinct digits (0–9) to the letters S, E, N, D, M, O, R, and Y.
4. Check that the leading digits S and M are not zero.
5. Convert SEND, MORE, and MONEY into their corresponding numerical values.
6. Check whether the equation **SEND + MORE = MONEY** is satisfied.
7. If the equation is satisfied, display the values of SEND, MORE, and MONEY.
8. Stop the program.

py4.py - C:/Users/navit/py4.py (3.12.10)

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```
from itertools import permutations

def solve_cryptarithmetic():
    for p in permutations(range(10), 8):
        S, E, N, D, M, O, R, Y = p

        if S == 0 or M == 0:
            continue

        send = 1000*S + 100*E + 10*N + D
        more = 1000*M + 100*O + 10*R + E
        money = 10000*M + 1000*O + 100*N + 10*E + Y

        if send + more == money:
            print("Solution Found!")
            print("SEND =", send)
            print("MORE =", more)
            print("MONEY =", money)
            return

    print("No solution found.")

solve_cryptarithmetic()
```

IDLE Shell 3.12.10

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```
Python 3.12.10 (tags/v3.12.10:0cc8128, Apr  8 2025, 12:21:36) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
========== RESTART: C:/Users/navit/py4.py ==========
Solution Found!
SEND = 9567
MORE = 1085
MONEY = 10652
>>>
```

Result:

Thus, the Python program successfully solves the cryptarithmetic problem and verifies that **9567 + 1085 = 10652**.